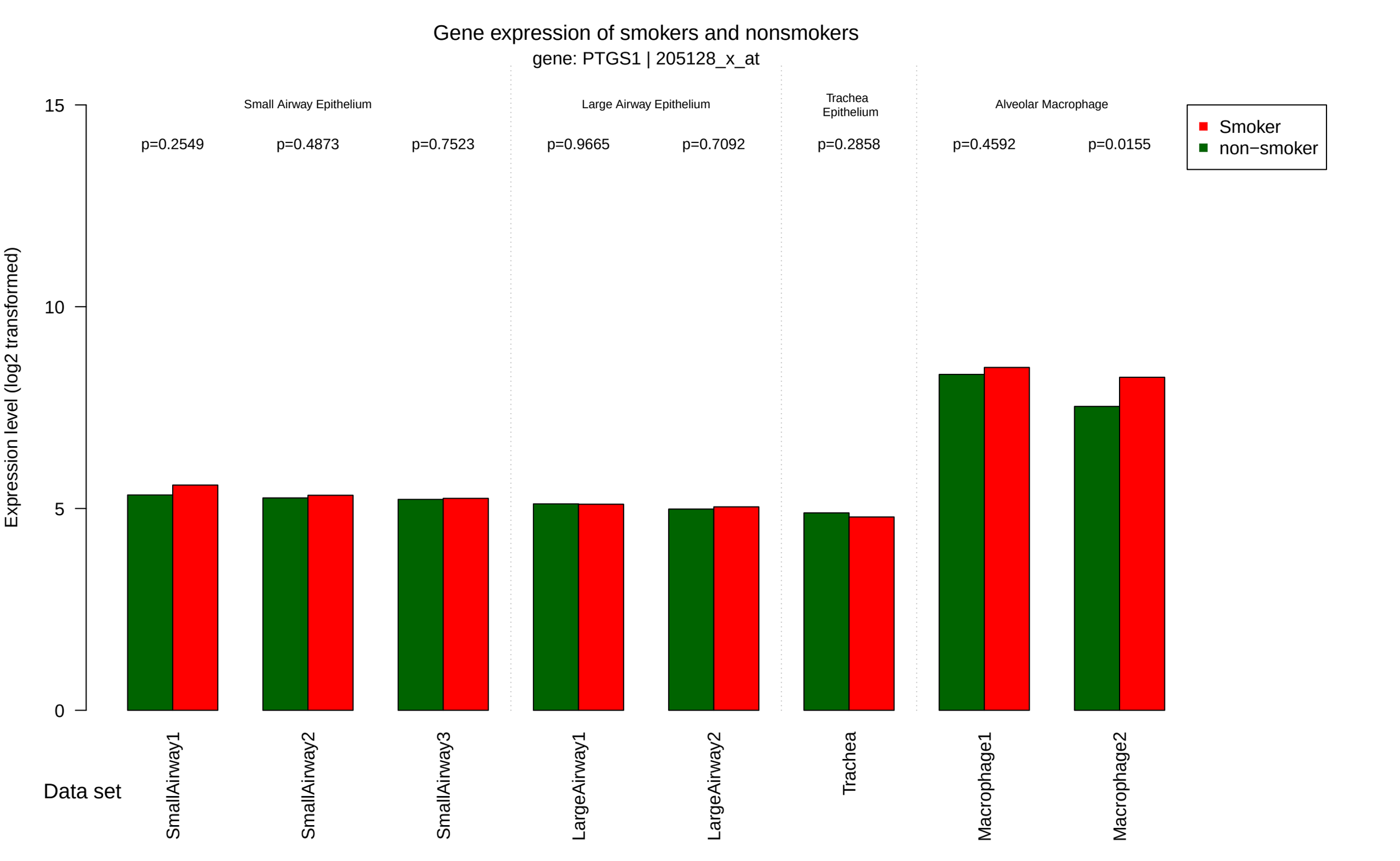
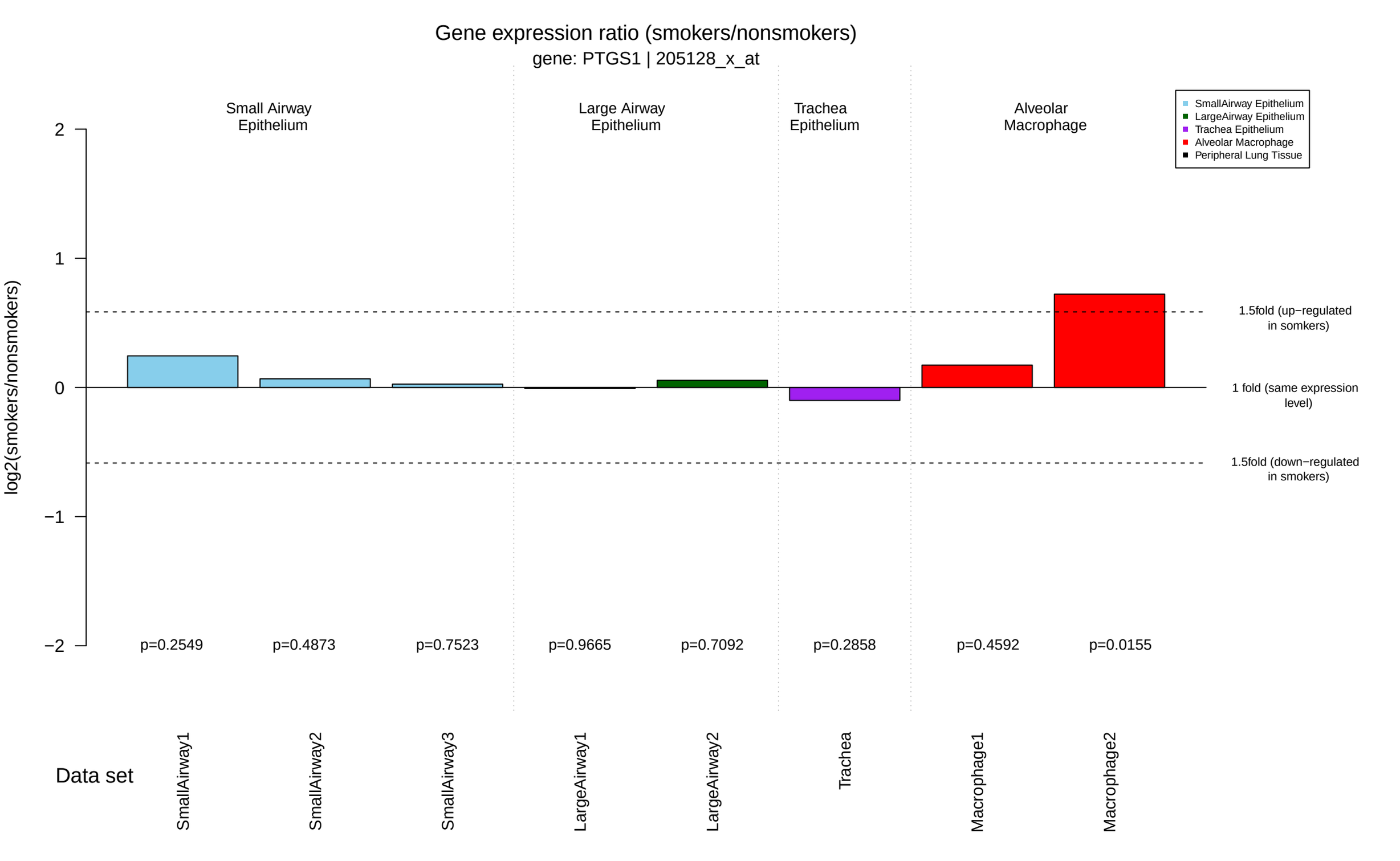
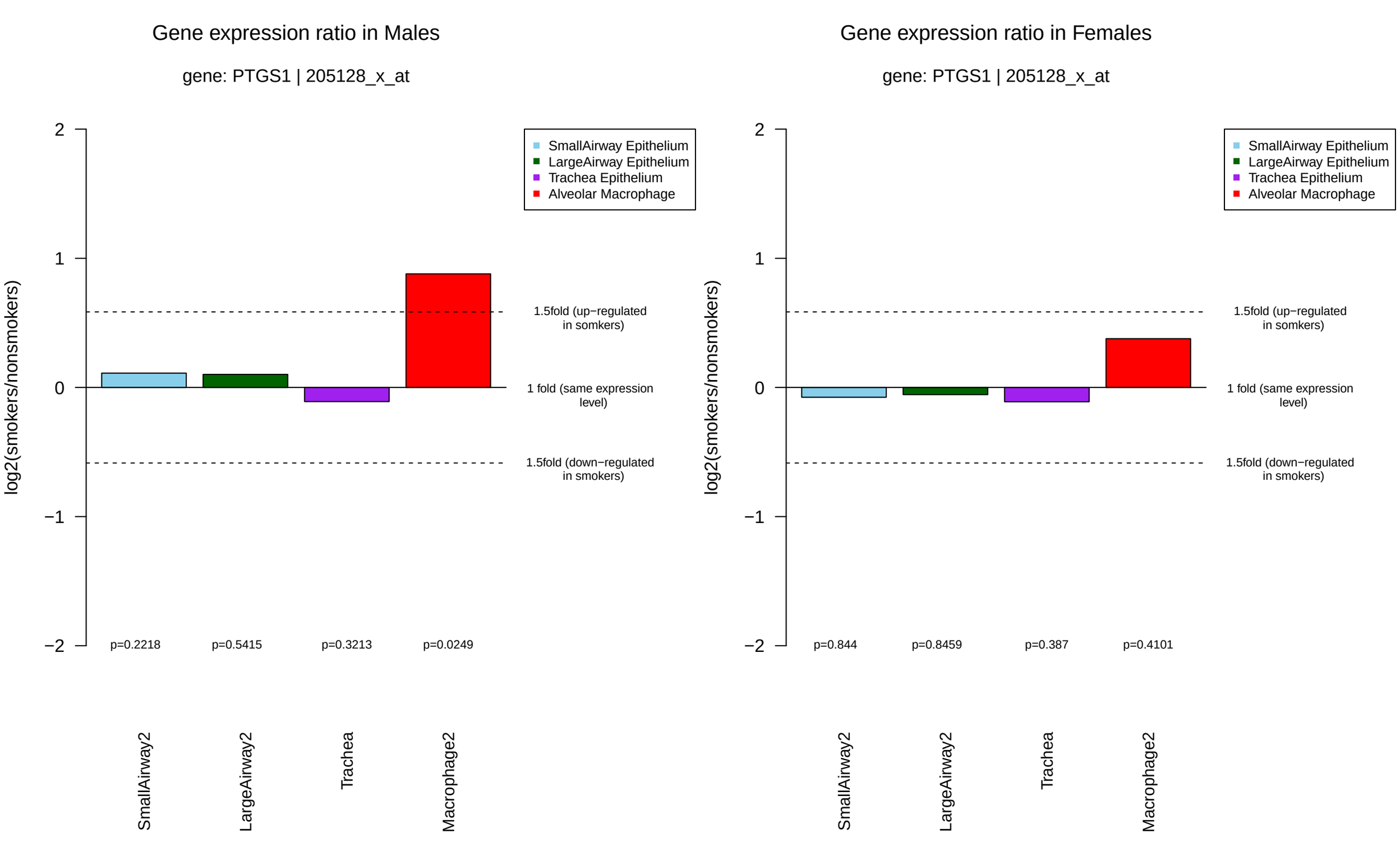
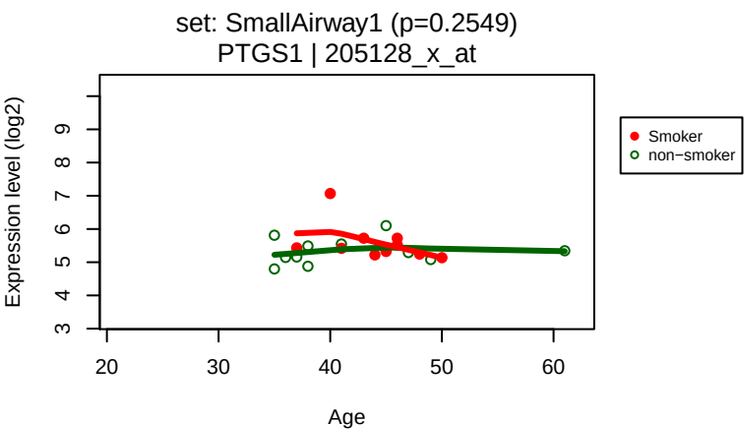
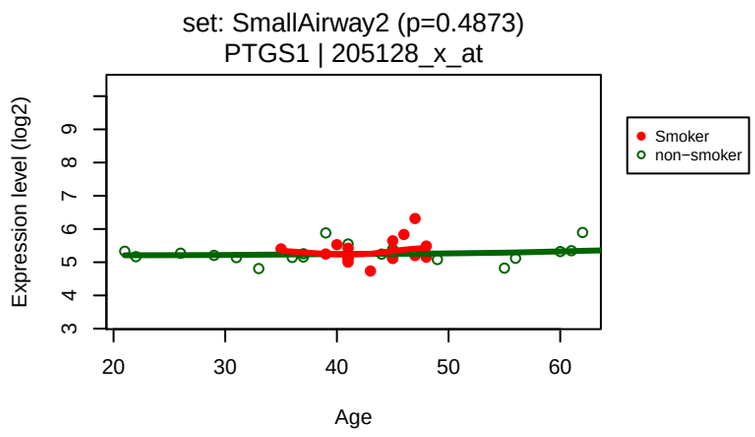
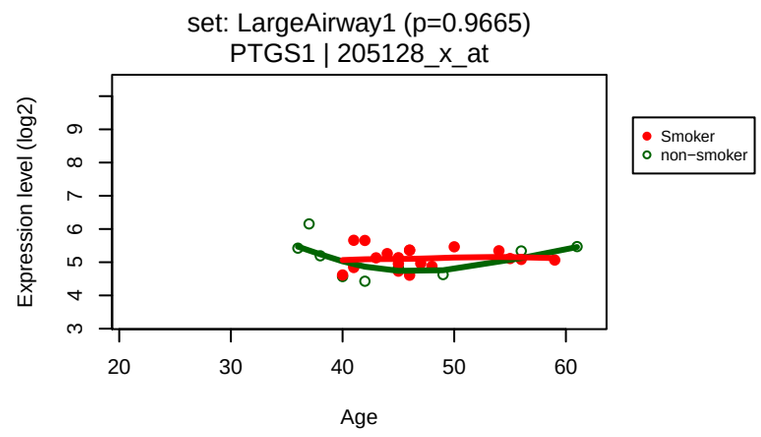

Gene expression of smokers and nonsmokers
gene: PTGS1 | 205128_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2549
p=0.4873
p=0.7523
p=0.9665
p=0.7092
p=0.2858
p=0.4592
p=0.0155
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

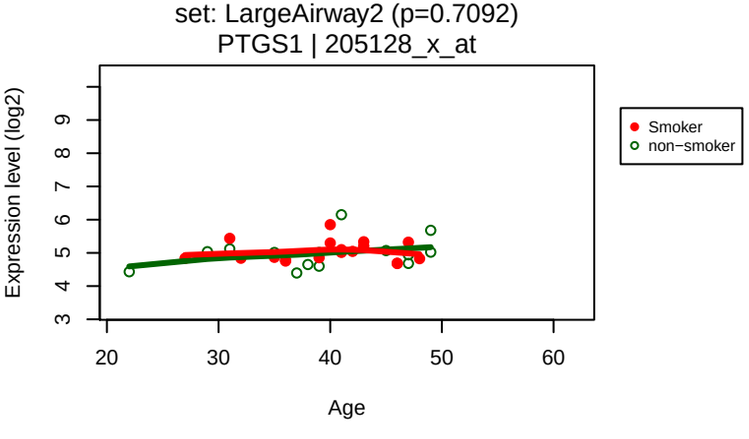

Gene expression ratio (smokers/nonsmokers)
gene: PTGS1 | 205128_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2549
p=0.4873
p=0.7523
p=0.9665
p=0.7092
p=0.2858
p=0.4592
p=0.0155
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

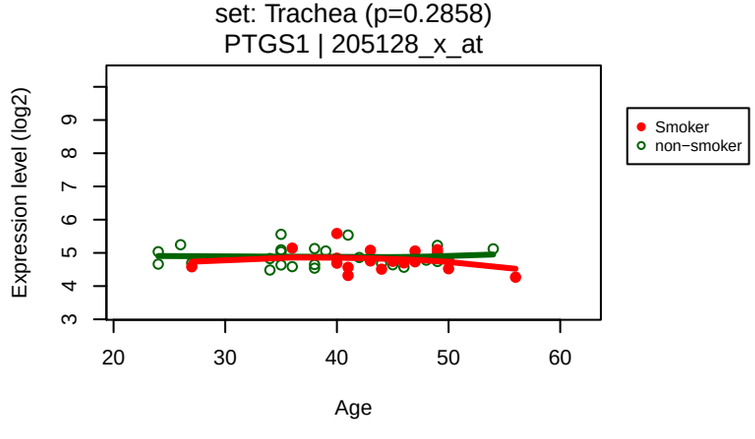

Gene expression ratio in Males
gene: PTGS1 | 205128_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2218
p=0.5415
p=0.3213
p=0.0249
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTGS1 | 205128_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.844
p=0.8459
p=0.387
p=0.4101
SmallAirway2
LargeAirway2
Trachea
Macrophage2

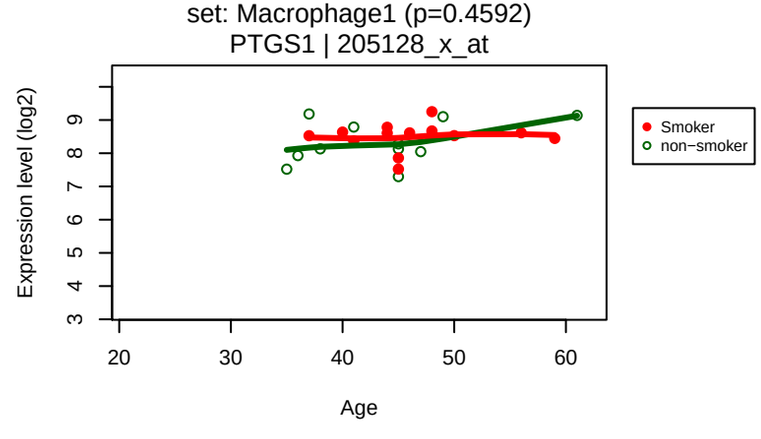

set: SmallAirway1 (p=0.2549)
PTGS1 | 205128_x_at
Expression level (log2)
Age
Smoker
non−smoker

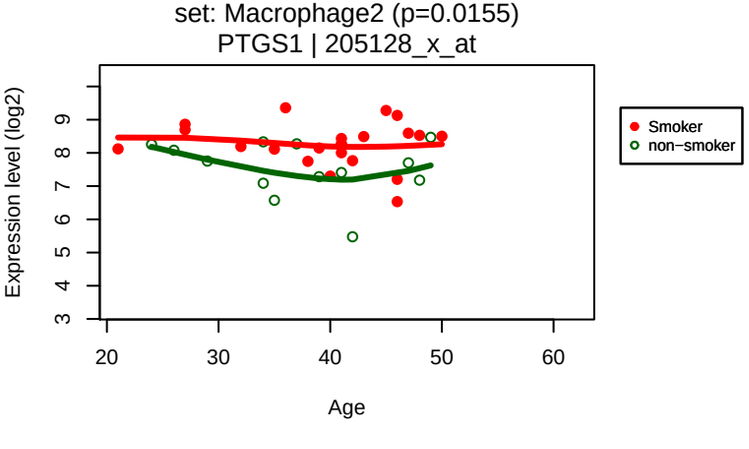

set: SmallAirway2 (p=0.4873)
PTGS1 | 205128_x_at
Expression level (log2)
Age
Smoker
non−smoker

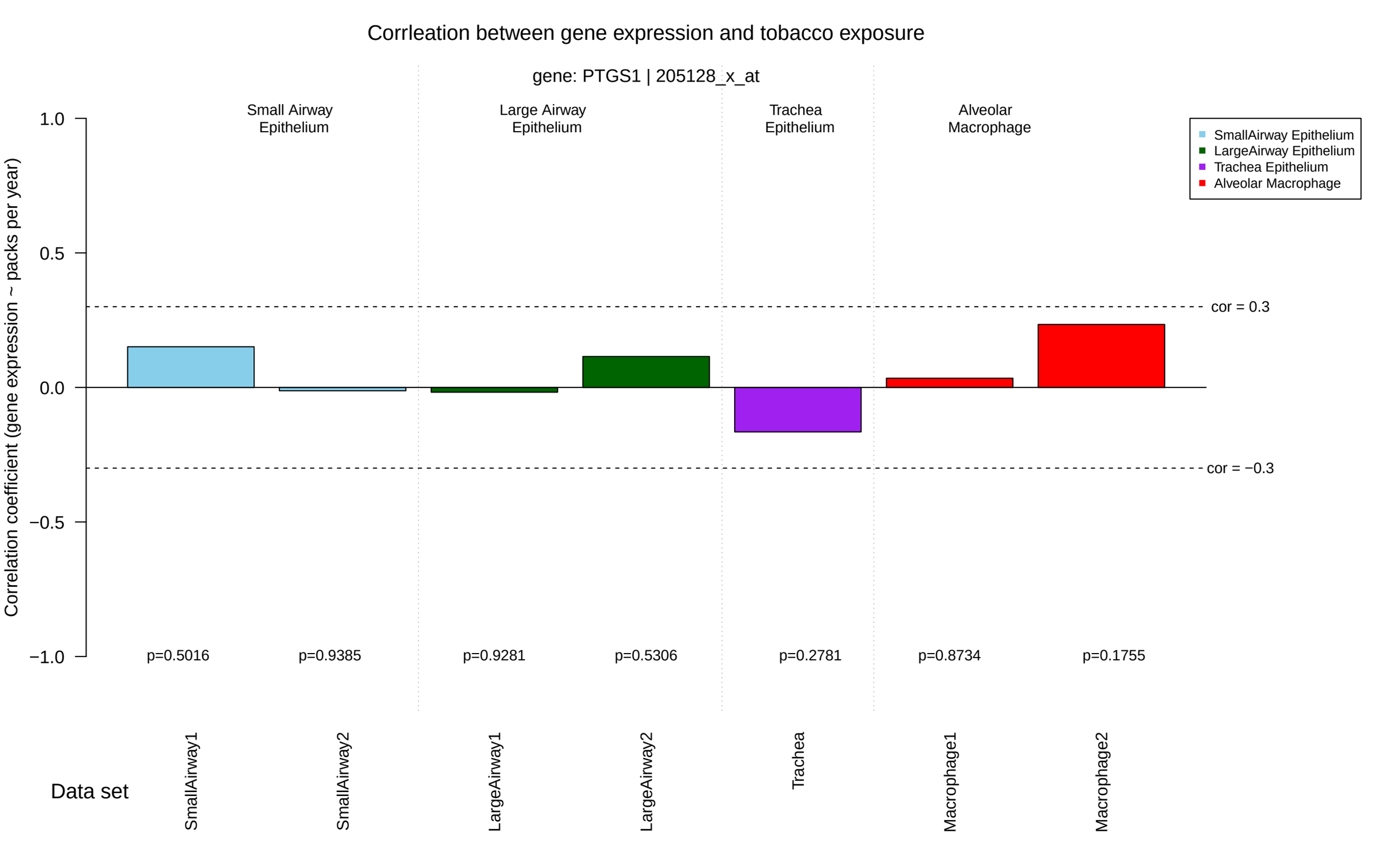

set: LargeAirway1 (p=0.9665)
PTGS1 | 205128_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7092)
PTGS1 | 205128_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2858)
PTGS1 | 205128_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4592)
PTGS1 | 205128_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0155)
PTGS1 | 205128_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTGS1 | 205128_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5016
p=0.9385
p=0.9281
p=0.5306
p=0.2781
p=0.8734
p=0.1755
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2